Kernel methods

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Generalized linear models

Typically,
$\phi_0(\mathbf{x}) = 1$ (bias term)

$$h(\mathbf{x};\, \boldsymbol{w}) = \boldsymbol{w}^T \boldsymbol{\phi}(\mathbf{x}) = w_0\phi_0(\mathbf{x}) + w_1\phi_1(\mathbf{x}) + \cdots + w_M\phi_M(\mathbf{x})$$

$$w_0\phi_0(\mathbf{x}_1) + w_2\phi_1(\mathbf{x}_1) + \cdots + w_M\phi_M(\mathbf{x}_1) = y_1$$

$$w_0\phi_0(\mathbf{x}_2) + w_2\phi_1(\mathbf{x}_2) + \cdots + w_M\phi_M(\mathbf{x}_2) = y_2$$

$$\cdots$$

$$w_0\phi_0(\mathbf{x}_N) + w_2\phi_1(\mathbf{x}_N) + \cdots + w_M\phi_M(\mathbf{x}_N) = y_N$$

Design matrix
$\Phi_{nm} = \phi_m(\mathbf{x}_n)$
$n = 1, \ldots, N$
$m = 0,1, \ldots, M$

$$\Rightarrow \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_M(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_M(\mathbf{x}_2) \\ \vdots & & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_M(\mathbf{x}_N) \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_M \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}$$

$$\boldsymbol{\Phi}\, \boldsymbol{w} = \mathbf{y} \quad \Rightarrow \quad \boldsymbol{\Phi}^T \boldsymbol{\Phi}\, \boldsymbol{w}^* = \boldsymbol{\Phi}^T \mathbf{y} \Rightarrow \qquad \boldsymbol{w}^* = (\boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{y}$$

Generalized linear models: Bayesian approach

Generative model: $y_n = \boldsymbol{w}^T \boldsymbol{\phi}(\mathbf{x}_n) + \epsilon_n; \quad n = 1,2,\dots,N$

Noise distribution: $\epsilon_n \sim N(0, \sigma_\epsilon)$

Prior distribution of weights: $\mathrm{p}(\boldsymbol{w}) = N(\boldsymbol{w} \mid \mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{w}})$

- Likelihood: $\mathrm{p}(\mathcal{D}|\boldsymbol{w}) \propto \frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \exp\left(-\frac{1}{2\sigma_\epsilon^2}\sum_{n=1}^{N}\left(y_n - \boldsymbol{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\right)^2\right)$

ML (maximum likelihood): $\boldsymbol{w}_{ML}^* = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{y}$

$(\boldsymbol{\Phi}_{test})_{nm} = \phi_m\left(\mathbf{x}_n^{(test)}\right)$
$n = 1,\dots,N_{test}$
$m = 0,1,\dots,M-1$

- Posterior distribution of weights

$\mathrm{p}(\boldsymbol{w}|\mathcal{D}) = N(\boldsymbol{w} \mid (\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \sigma_\epsilon^2\boldsymbol{\Sigma}_{\boldsymbol{w}}^{-1})^{-1}\boldsymbol{\Phi}^T\mathbf{y}, (\sigma_\epsilon^{-2}\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \boldsymbol{\Sigma}_{\boldsymbol{w}}^{-1})^{-1})$

MAP (maximum posterior): $\boldsymbol{w}_{MAP}^* = (\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \sigma_\epsilon^2\boldsymbol{\Sigma}_{\boldsymbol{w}}^{-1})^{-1}\boldsymbol{\Phi}^T\mathbf{y}$

- Predictive distribution: $\boldsymbol{A} = \sigma^{-2}\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \boldsymbol{\Sigma}_{\boldsymbol{w}}^{-1};$

$$\mathrm{p}(\boldsymbol{y}_{test}|\mathbf{X}_{test}, \mathcal{D}) = N(\boldsymbol{y}_{test}|\sigma_\epsilon^{-2}\boldsymbol{\Phi}_{test}\boldsymbol{A}^{-1}\boldsymbol{\Phi}^T\mathbf{y}, \boldsymbol{\Phi}_{test}\boldsymbol{A}^{-1}\boldsymbol{\Phi}_{test}')$$

Bayesian approach and regularization

- Prior for weights: Prior distribution of weights: $\mathrm{p}(\boldsymbol{w}) = N(\boldsymbol{w} \mid \mathbf{0}, \sigma_{\boldsymbol{w}}^2 \mathbf{I})$
- osterior distribution of weights

$$\mathrm{p}(\boldsymbol{w}|\mathcal{D}) = N\left(\boldsymbol{w} \;\middle|\; \left(\boldsymbol{\Phi}^T \boldsymbol{\Phi} + \frac{\sigma_\epsilon^2}{\sigma_{\boldsymbol{w}}^2}\mathbf{I}\right)^{-1} \boldsymbol{\Phi}^T \mathbf{y}, \sigma_\epsilon^{-2}\left(\boldsymbol{\Phi}^T \boldsymbol{\Phi} + \frac{\sigma_\epsilon^2}{\sigma_{\boldsymbol{w}}^2}\mathbf{I}\right)^{-1}\right)$$

MAP (maximum posterior): $\boldsymbol{w}_{MAP}^* = \left(\boldsymbol{\Phi}^T \boldsymbol{\Phi} + \frac{\sigma_\epsilon^2}{\sigma_{\boldsymbol{w}}^2}\mathbf{I}\right)^{-1} \boldsymbol{\Phi}^T \mathbf{y}$

Equivalent to kernel ridge regression

$$\boldsymbol{w}_{MAP}^* = \arg\min_{\boldsymbol{w}} \left[\sum_{n=1}^{N} \left(y_n - \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_n)\right)^2 + \frac{\sigma_\epsilon^2}{\sigma_{\boldsymbol{w}}^2} \boldsymbol{w}^T \boldsymbol{w}\right]$$

Least squares + L_2 regularization (ridge regression) in feature space

Sum of squared residuals

L_2 penalty

Gaussian processes

Other assumptions for the noise distribution possible

Generative model: $y_n = f(\mathbf{x}_n) + \epsilon_n; \qquad n = 1,2,\dots,N$

Noise distribution: $\epsilon_n \sim N(0, \sigma_\epsilon^2)$

Often, $m(\mathbf{x}) = 0$ or $m(\mathbf{x}) = const$

Prior distribution of $f(\mathbf{x})$: $f(\mathbf{x}) \sim GP\big(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')\big)$

$$m(\mathbf{x}) = \mathrm{E}[f(\mathbf{x})]; \quad k(\mathbf{x}, \mathbf{x}') = \mathrm{E}\big[\big(f(\mathbf{x}) - m(\mathbf{x})\big)\big(f(\mathbf{x}') - m(\mathbf{x}')\big)\big]$$

- Predictive distribution: $\mathrm{p}(\boldsymbol{y}_{test} | \mathbf{X}_{test}, \mathcal{D}) = N\big(\boldsymbol{y}_{test} | \bar{\boldsymbol{y}}_{test}, cov(\boldsymbol{y}_{test})\big)$

$$\bar{\boldsymbol{y}}_{test} = \boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X})(\boldsymbol{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})^{-1}\, \boldsymbol{y}$$

$$cov(\boldsymbol{y}_{test}) = (\boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X}_{test}) + \sigma_\epsilon^2 \mathbf{I}) - \boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X})(\boldsymbol{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})^{-1}\, \boldsymbol{K}(\mathbf{X}, \mathbf{X}_{test})$$

$$\big(\boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X}_{test})\big)_{ln} = k\left(x_l^{(test)}, x_n^{(test)}\right); \quad l, n = 1,2,\dots,N_{test}$$

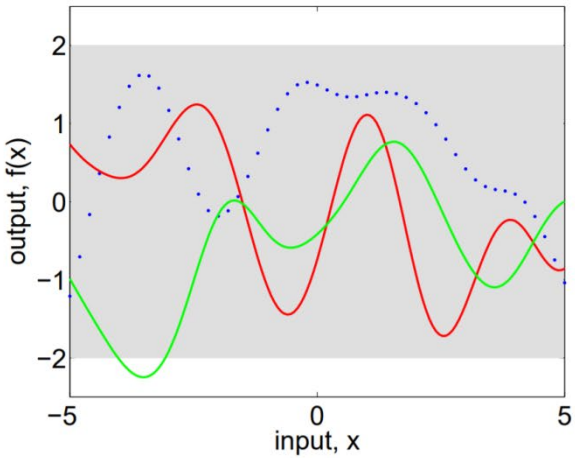
$$\big(\boldsymbol{K}(\mathbf{X}, \mathbf{X})\big)_{ij} = k(\mathbf{x}_i, \mathbf{x}_j); \quad i, j = 1,2,\dots,N$$

$$\big(\boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X})\big)_{lj} = k\left(x_l^{(test)}, \mathbf{x}_j\right);$$

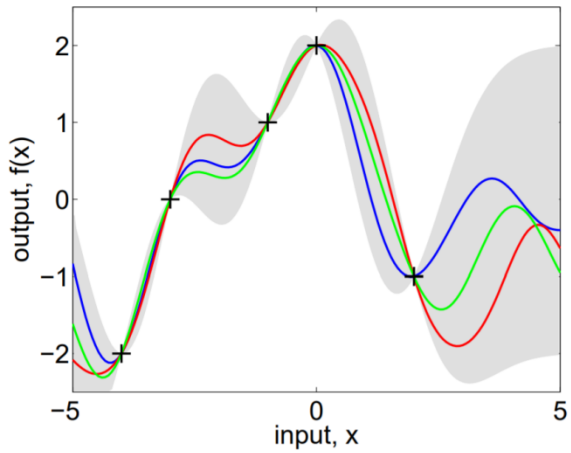
$$\big(\boldsymbol{K}(\mathbf{X}, \mathbf{X}_{test})\big)_{in} = k\left(\mathbf{x}_i, x_n^{(test)}\right);$$

$$\boldsymbol{K}(\mathbf{X}, \mathbf{X}) = \boldsymbol{\Phi}\boldsymbol{\Sigma}_{\boldsymbol{w}}\, \boldsymbol{\Phi}^T$$
$$\boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X}) = \boldsymbol{\Phi}_{test}\, \boldsymbol{\Sigma}_{\boldsymbol{w}} \boldsymbol{\Phi}^T$$
$$\boldsymbol{K}(\mathbf{X}_{test}, \mathbf{X}_{test}) = \boldsymbol{\Phi}_{test}\, \boldsymbol{\Sigma}_{\boldsymbol{w}} \boldsymbol{\Phi}_{test}^T$$

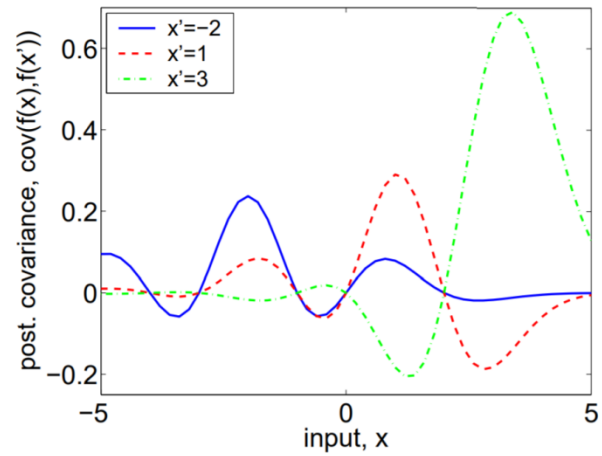
Gaussian processes for regression



(a), prior



(a), posterior



(b), posterior covariance

Figures from C. E. Rasmussen & C. K. I. Williams, Gaussian Processes for Machine Learning, the MIT Press, 2006

Kernel functions: Properties

$k(\mathbf{x}, \mathbf{x}')$: Symmetric, positive definite function

- New kernels from $k_1(\mathbf{x}, \mathbf{x}')$, $k_2(\mathbf{x}, \mathbf{x}')$, kernels
  - $k_1(\mathbf{x}, \mathbf{x}') + \sigma_1^2$
  - $\sigma_1^2 k_1(\mathbf{x}, \mathbf{x}')$
  - $k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$
  - $k_1(\mathbf{x}, \mathbf{x}') k_2(\mathbf{x}, \mathbf{x}')$

- Simple kernels
  - Linear (homogeneous): $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}$
  - Linear (inhomogeneous): $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^T \mathbf{x}$
  - Linear (inhomogeneous, fully parametrized):

$$k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \sigma_1^2 (\mathbf{x} - \mathbf{c})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \mathbf{c})$$

  - Polynomial (degree p): $k(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^T \mathbf{x})^p$

Kernel functions

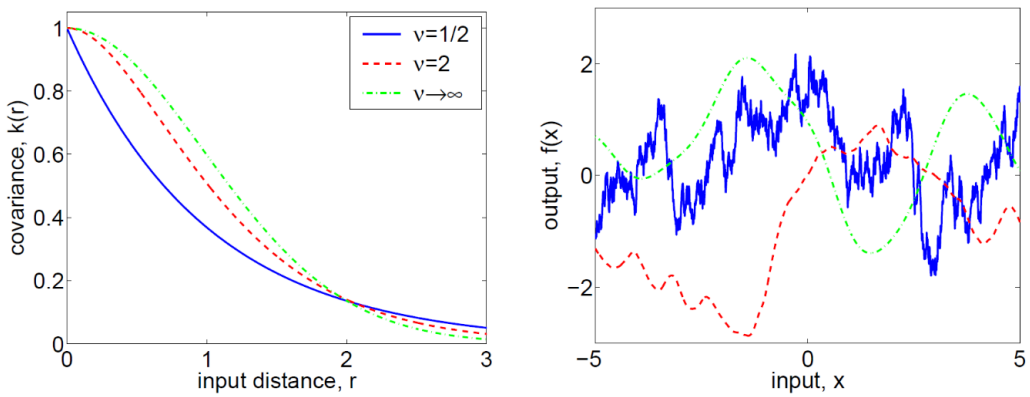
- Gaussian, RBF, squared exponential (SE)

$$k(\mathbf{x},\mathbf{x}') = \sigma^2 exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2s^2}\right)$$

Very smooth (in green)

- Exponential: $k(\mathbf{x},\mathbf{x}') = \sigma^2 exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|_1}{s}\right)$

Very rough (in blue)



(a) (b)

Figure 4.1: Panel (a): covariance functions, and (b): random functions drawn from Gaussian processes with Matérn covariance functions, eq. (4.14), for different values of ν, with $\ell = 1$. The sample functions on the right were obtained using a discretization of the x-axis of 2000 equally-spaced points.

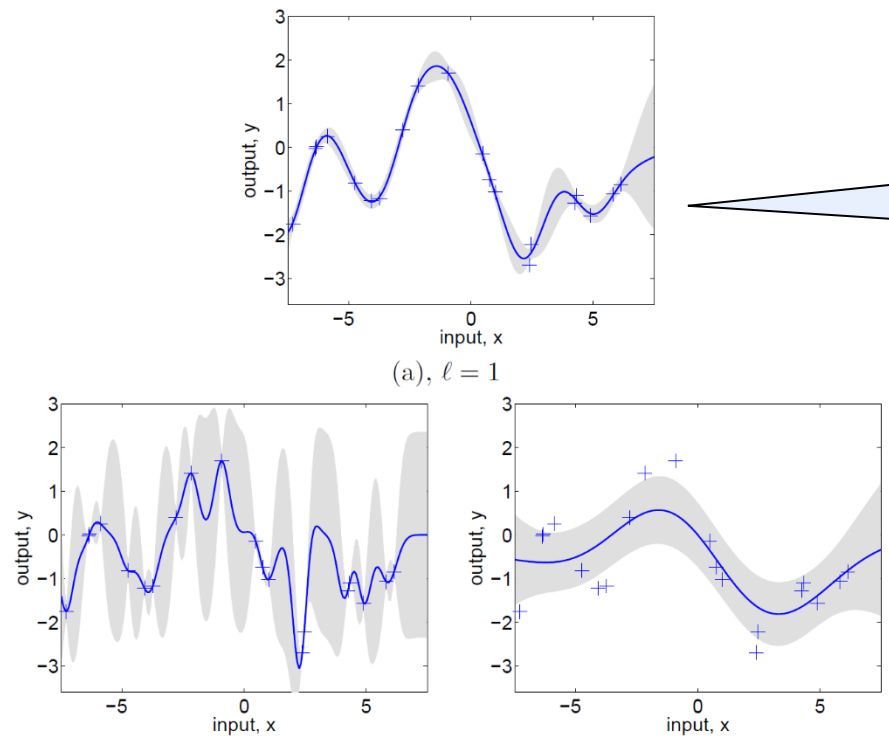
Figure from
C. E. Rasmussen & C. K. I. Williams,
Gaussian Processes for Machine Learning
The MIT Press, 2006

Other kernel functions

- Gamma exponential: $k(\mathbf{x},\mathbf{x}') = \sigma^2 exp\left(-\left(\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}\right)^{\gamma}\right); 0 < \gamma \leq 2$
- Matérn: $k(\mathbf{x},\mathbf{x}') = \sigma^2 \frac{2^{1-\upsilon}}{\Gamma(\upsilon)}\left(\frac{\sqrt{2\upsilon}}{s}\|\mathbf{x}-\mathbf{x}'\|\right)^{\upsilon} K_{\upsilon}\left(\frac{\sqrt{2\upsilon}}{s}\|\mathbf{x}-\mathbf{x}'\|\right); 0 < \upsilon \leq \infty$
  - Exponential with L^2 norm $\upsilon = 1/2$
  - Squared exponential $\upsilon = \infty$
- Matérn $\upsilon = 3/2$: $k(\mathbf{x},\mathbf{x}') = \sigma^2\left(1+\sqrt{3}\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}\right) exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}\right)$
- Matérn $\upsilon = 5/2$: $k(\mathbf{x},\mathbf{x}') = \sigma^2\left(1+\sqrt{5}\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}+\frac{5}{3}\left(\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}\right)^2\right) exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|}{s}\right)$
- Periodic kernel: $k(\mathbf{x},\mathbf{x}') = \sigma^2 exp\left(-\frac{2sin^2\frac{\pi\|\mathbf{x}-\mathbf{x}'\|}{p}}{2s^2}\right)$

Reasonable choice (in general)

Dependence on the kernel (hyper-)parameters



(a), $\ell = 1$

(b), $\ell = 0.3$

(c), $\ell = 3$

Figure 2.5: (a) Data is generated from a GP with hyperparameters $(\ell, \sigma_f, \sigma_n) = (1, 1, 0.1)$, as shown by the + symbols. Using Gaussian process prediction with these hyperparameters we obtain a 95% confidence region for the underlying function f (shown in grey). Panels (b) and (c) again show the 95% confidence region, but this time for hyperparameter values $(0.3, 1.08, 0.00005)$ and $(3.0, 1.16, 0.89)$ respectively.

Figure from
C. E. Rasmussen & C. K. I. Williams,
Gaussian Processes for Machine Learning
The MIT Press, 2006

Learning the kernel (hyper-)parameters

- Determine the kernel parameters by cross-validation (costly!)

- Maximize the marginal (type II) likelihood with respect to the kernel parameters

$$\log \mathrm{p}(\boldsymbol{y}|\mathbf{X}) = -\frac{1}{2}\boldsymbol{y}^T(\boldsymbol{K}(\mathbf{X},\mathbf{X}) + \sigma_\epsilon^2\mathbf{I})\boldsymbol{y} - \frac{1}{2}log|\boldsymbol{K}(\mathbf{X},\mathbf{X}) + \sigma_\epsilon^2\mathbf{I}| - \frac{N}{2}\log 2\pi$$

Error term

Complexity penalty

  - More complex (possibly numerical) treatment is required when noise is not Gaussian (doable!)

Support Vector Machine (SVM)

Reproducing Kernel Hilbert Space

Input: $\mathcal{D}_{train} = \{(\mathbf{x}_n, c_n)\}_{n=1}^{N_{train}}$

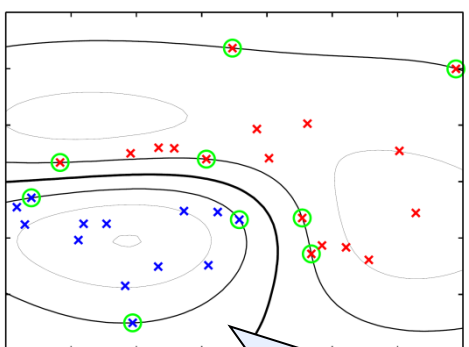
Kernel: $K(\mathbf{x}, \mathbf{x}'; \gamma)$ [inner product in RKHS]

Parameter for regularization: $C > 0$

Salida: **Support vectors (SVs)**

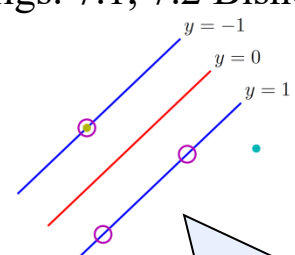
Código:

1. Formulate the problem in RKHS associated to $K(\mathbf{x}, \mathbf{x}'; \boldsymbol{\sigma})$
2. Find the **optimal separating hyperplane** (**maximize margin**) solving a **convex optimization** problem.

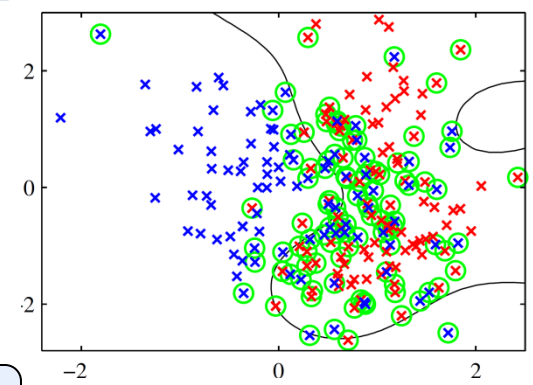


Original space

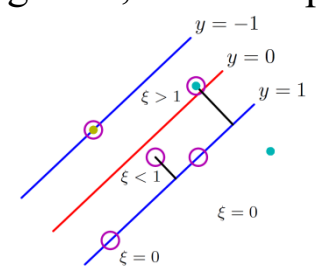
Separable case
(figs. 7.1, 7.2 Bishop)



Space of features



Non-separable case
(figs. 7.3, 7.4 Bishop)



SVM: Maximizing the margin

- ± 1 encoding of the **labeled data**

$$\mathcal{D} = \{(\mathbf{x}_n, c_n)\}_{n=1}^{N} \Rightarrow \mathcal{D} = \{(\mathbf{x}_n, t_n)\}_{n=1}^{N}; \;\; t_n = \begin{cases} -1 & \text{if} \;\; c_n = C_0 \\ 1 & \text{if} \;\; c_n = C_1 \end{cases}$$

- Linear model in feature space:

$$h(\mathbf{x}; \mathbf{w}) = w_0 + \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x})$$

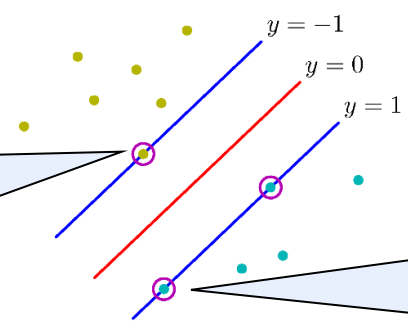
- Maximization of the margin

$$\mathbf{w}^* = \underset{w_0, \mathbf{w},}{\arg\min} \left(\frac{1}{2} \mathbf{w}^{\mathrm{T}} \mathbf{w} \right);$$

Convex optimization (unique solution)

$$\text{such that} \;\; \left(w_0 + \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) \right) t_n \geq 1$$

(figs. 7.1, Bishop)



Support Vector
$\left(\mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) + b \right) t_n = 1$
$t_n = -1$

Support Vector
$\left(\mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) + b \right) t_n = 1$
$t_n = 1$

Non-linearly separable clase: slack variables

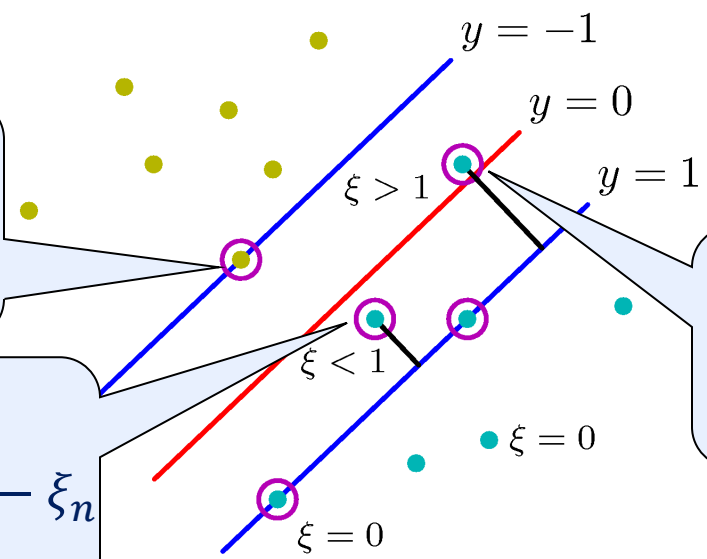
- Maximization of the margin

$$\mathbf{w}^* = \arg\min_{\mathbf{w},b} \left(\frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w} + C\sum_{n=1}^{N} \xi_n \right),$$

$C > 0$
In the limit $C \to \infty$ one gets the linearly separable clase.

$$\text{such that } \left(w_0 + \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(\mathbf{x}_n)\right) t_n \geq 1 - \xi_n; \quad \xi_n \geq 0$$

(fig. 7.3, Bishop)



Support Vector
$\left(w_0 + \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(\mathbf{x}_n)\right) t_n = 1$
$t_n = -1; \quad \xi_n = 0$

Support Vector
$\left(w_0 + \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(\mathbf{x}_n)\right) t_n = 1 - \xi_n$
$t_n = -1; \quad 0 < \xi_n < 1$

Support Vector
$\left(w_0 + \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(\mathbf{x}_n)\right) t_n = 1$
$t_n = 1; \quad \xi_n > 1$

Kernel idea: Embedding in feature space

- **Original space**: $\mathbf{x}^{\mathrm{T}} = (x_1 \quad x_2)$
- **Feature space**: $\boldsymbol{\phi}^{\mathrm{T}}(\mathbf{x}) = \left(c \quad \sqrt{2c\gamma}x_1 \quad \sqrt{2c\gamma}x_2 \quad \gamma x_1^2 \quad \sqrt{2}\gamma x_1 x_2 \quad \gamma x_2^2\right)$

$c \geq 0$

$\gamma \geq 0$

- **Quadratic kernel:**

$$K(\mathbf{x}_m, \mathbf{x}_n) = \langle \boldsymbol{\phi}(\mathbf{x}_m) | \boldsymbol{\phi}(\mathbf{x}_n) \rangle = (c + \gamma \mathbf{x}_m^T \mathbf{x}_n)^2 = c^2 + 2c\gamma \mathbf{x}_m^T \mathbf{x}_n + \gamma^2 (\mathbf{x}_m^T \mathbf{x}_n)^2$$
$$= c^2 + 2c\gamma x_{m1} x_{n1} + 2c\gamma x_{m2} x_{n2}$$
$$+ \gamma^2 x_{m1}^2 x_{n1}^2 + 2\gamma^2 x_{m1} x_{m2} x_{n1} x_{n2} + \gamma^2 x_{m2}^2 x_{n2}^2$$

SVM + linear kernel can be trained fast!

Equivalently: Use SVM + linear kernel with extended fearures

$$\boldsymbol{\phi}^{\mathrm{T}}(\mathbf{x}) = (1 \quad x_1 \quad x_2 \quad x_1^2 \quad x_1 x_2 \quad x_2^2)$$

Chang, Yin-Wen; Hsieh, Cho-Jui; Chang, Kai-Wei; Ringgaard, Michael; Lin, Chih-Jen (2010). "Training and testing low-degree polynomial data mappings via linear SVM". Journal of Machine Learning Research. 11: 1471–1490.

SVM design

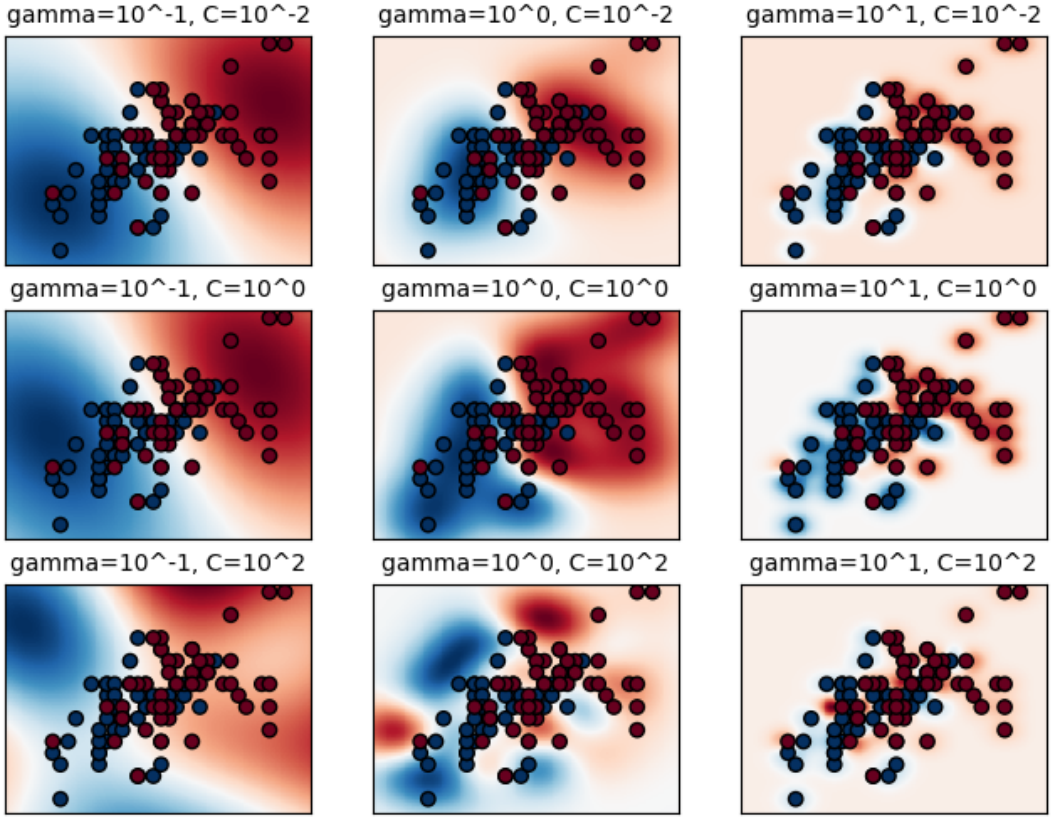
- **Kernel:** $K(\mathbf{x}, \mathbf{x}'; \gamma)$
  - Linear, polynomial
  - Squared exponential (a.k.a. Gaussian, RBF)
  - Sigmoid
  - Kernels for strings, texts, images, etc.
- **Kernel inverse scale:** $\gamma \in \{2^{-15}, 2^{-13}, \ldots, 2^{3}\}$
  - High γ (narrower kernel): More flexibility (can overfit)
  - Low γ (broader kernel): More robustness (can underfit)
- **Regularization strength (penalty of slack variables)** $C \in \{2^{-5}, 2^{-3}, \ldots, 2^{15}\}$
  - High C (less regularization. Fewer SVs): More flexibility (can overfit)
  - Low C (more regularization. More SVs): More robustness (can underfit)
- **In SVR:** ε-insensitive hinge loss: ε ($\approx$ level of noise. Lower with more data)
  - Low ε (less regularization. More SVs): More flexibility (can overfit)
  - high ε (more regularization. Fewer SVs): More robustness (can underfit)

If components of attribute vector are in [0,1] , use

$$\gamma = \frac{2^{-k}}{D};\ k = 0, 1, 2, \ldots$$

Alternative: Control the number of support vectors Using ν-SVM

SVM: Kernel parameters



http://scikit-learn.org/stable/_images/sphx_glr_plot_rbf_parameters_001.png

SVM summary

- **Advantages**
  - Excellent predictors.
  - Solid theoretical foundation (*structural risk minimization*).
  - Learning = convex optimization (unique solution).
- **Disadvantages**
  - Costly training.
  - Original formulation for binary classification problems only.
  - Finding appropriate values of hyperparameters can be difficult:
    - Kernel type
    - Kernel parameters: $\gamma > 0$
    - Strength of regularization term $C > 0$
  - Difficult interpretation.
  - Non-adaptive.

Can be extended to multiple classes!

State-of-the-art: SVM, SVR

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